

ROLL NO:250701573

EXPERIMENT NO: 12 CONDUCT Z TEST FOR THE DATA GIVEN USING PYTHON

PROGRAM: import numpy as np

import scipy.stats as stats

Define the sample data (changed bird weights in grams)

```
sample_data = np.array([
    157, 153, 156, 154, 152, 158, 155, 153, 157, 154,
    156, 155, 154, 157, 156, 153, 155, 157, 154, 155,
    153, 158, 156, 155, 154, 157, 153, 156, 155, 158
])
```

Population mean under the null hypothesis

population_mean = 155

Calculate sample statistics

sample_mean = np.mean(sample_data)

sample_std = np.std(sample_data, ddof=1)

sample_size = len(sample_data)

Calculate Z-score

z_score = (sample_mean - population_mean) / (sample_std / np.sqrt(sample_size))

Calculate p-value

p_value = 2 * (1 - stats.norm.cdf(abs(z_score)))

```
# Significance level
alpha = 0.05

print("Sample Mean:", sample_mean)
print("Sample Standard Deviation:", sample_std)
print("Z-score:", z_score)
print("P-value:", p_value)

if p_value < alpha:
    print("Reject the Null Hypothesis")
else:
    print("Fail to Reject the Null Hypothesis")
```

OUTPUT:

```
Sample Mean: 155.2
Sample Standard Deviation: 1.7100110909097876
Z-score: 0.6406070234476979
P-value: 0.5217780352255008
Fail to Reject the Null Hypothesis
```